

Volatility Forecasting: GARCH and Recurrent Models

Error reduction, forecast validity, and temporal uncertainty

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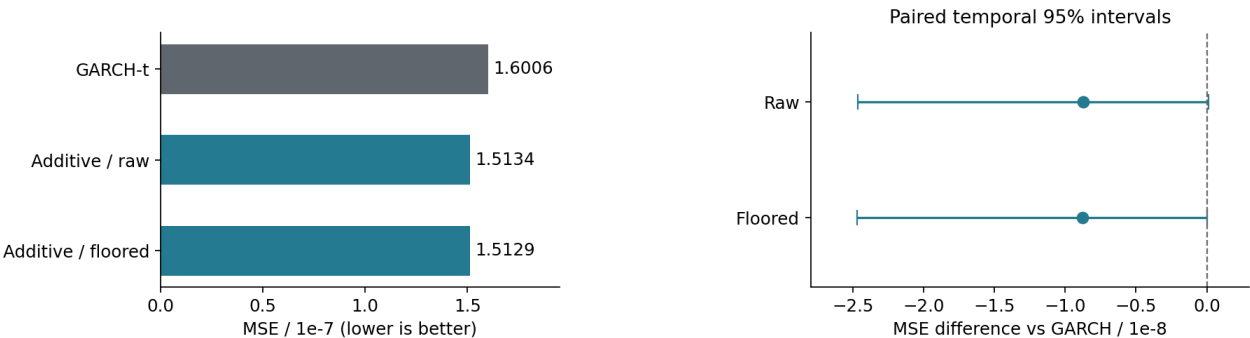
Executive Summary

An additive-residual LSTM achieves **5.48% lower observed MSE** than GARCH-t with a fixed output floor, and 5.45% lower without it. The comparison uses 3,738 identical daily targets across 178 walk-forward folds.

The result does not establish robust superiority: the primary 95% paired temporal interval includes zero. The raw model produces 60 nonpositive forecasts, and flooring them yields QLIKE of 457,801, versus -8.4613 for GARCH.

Additive-residual LSTM: error and uncertainty

3,738 common test dates | 2010-02-05 to 2024-12-11 | one training-seed schedule



Observed MSE change: -5.48% | both intervals include zero

60 nonpositive raw forecasts; floored QLIKE 457,801 vs GARCH -8.4613.

Historical study, not an untouched holdout. Intervals condition on trained forecasts; no seed or selection adjustment.

Source: reports/additive_results.json | paired stationary bootstrap, mean block 21, 2,000 resamples

Finding: aligning the training objective with variance MSE helps this point estimate, but does not guarantee positive, calibrated variance forecasts. This is a single-seed historical study, not evidence of trading alpha or a deployment-ready model.

Evaluation: 2010-02-05 to 2024-12-11. Source: committed benchmark and additive aggregate artifacts; no new training or market-data download was performed to assemble this report.

01 / Experiment Design

Component	Specification
Data and target	Cached S&P 500 index (^GSPC) daily-price snapshot, 2003-2024. Target y is squared daily log return, a noisy second-moment proxy. Its interpretation as conditional variance assumes a zero conditional mean.
Benchmarks	Student-t GARCH(1,1), a lagged 21-day mean of squared returns, and EWMA(0.94). GARCH parameters are refitted on expanding returns every 21 observations; the conditional-variance state updates each day.
Neural candidates	Pure and GARCH-feature LSTM/GRU models; multiplicative log-ratio LSTM/GRU models; and an additive-residual LSTM. Raw and floored additive forecasts are two outputs of the same fitted model.
Input availability	21 lagged rows ending at $t-1$: log return, squared return, absolute return, and annualized 21-day rolling volatility. Feature and residual hybrids add lagged GARCH variance. Targets remain in daily variance units.
Walk-forward splits	After a 756-return GARCH warmup, initial neural training contains 756 sequences from 777 rows. Validation has 252 observations; test blocks have 21. Training expands across 178 non-overlapping test blocks. The incomplete trailing block is excluded.
Training	Eight recurrent units; dropout 0.1; Adam 0.001; batch 64; at most 35 epochs; patience six; seed 42 plus split ID; best-validation checkpoint restored. CPU recurrence is unrolled for the fixed sequence length.

Information discipline

A forecast for t uses information through $t-1$. Feature and target transforms are fitted on each fold's training sample only. Validation chooses early stopping. A past test observation may enter a later fold's training history once it is observable.

Experiment scope

The additive specification was evaluated in a separate run after inspection of the historical sample. It uses the same dates, benchmark predictions, architecture, seed schedule, and training budget. The report combines the evidence, not the fits. Neither experiment is an untouched prospective holdout.

02 / Complete Same-Date Results

Forecast	MSE / 10 ⁻⁷	QLIKE	Mean h / mean y
GARCH-t	1.60062	-8.4613	1.038
Lagged RV21	1.87208	-8.3529	1.005
EWMA 0.94	1.74818	-8.4200	1.004
Pure LSTM	1.93876	-6.4483	0.258
Pure GRU	1.91925	-6.4303	0.259
Feature LSTM	1.96415	-6.6066	0.262
Feature GRU	1.97011	-6.5149	0.256
Ratio LSTM	1.86316	-6.6595	0.299
Ratio GRU	1.85128	-6.5718	0.283
Additive LSTM: raw	1.51343	Undefined	1.049
Additive LSTM: floored	1.51293	457,801.0548	1.051

Table 1. All rows use the same 3,738 squared-return targets. Lower MSE and QLIKE are better. The final two rows share one trained LSTM; the floor is 1e-12. Undefined raw QLIKE is not replaced by a favorable score. Mean h / mean y is a level-calibration diagnostic, not a test of conditional calibration.

How to interpret the ranking

GARCH-t beats all six log-target/log-ratio recurrent candidates on both MSE and QLIKE. The additive model has a lower MSE point estimate, but its raw outputs can be invalid variances. A favorable ordering on one loss is not an overall model verdict.

Scoring conventions

$$\text{MSE} = \text{mean}[(y - h)^2]$$

$$\text{QLIKE} = \text{mean}[\log(h) + y / h]$$

Reported QLIKE uses a 1e-12 numerical floor for positive-output evaluation, including zero targets. It can be negative. Raw additive QLIKE remains undefined because some forecasts are nonpositive. Calendar and uncertainty diagnostics use matched dates.

03 / Uncertainty and Stability

Additive minus GARCH	Difference	95% lower	95% upper
Raw MSE / 10^{-7}	-0.0871944	-0.246742	+0.00119215
Floored MSE / 10^{-7}	-0.0876892	-0.247306	+0.000338485
Floored QLIKE	+457810	+239583	+754554

Table 2. Negative differences favor the additive model. Paired stationary-bootstrap 95% percentile intervals use mean block length 21 and 2,000 replications. Both MSE intervals include zero; these are pointwise temporal intervals conditional on the trained forecasts, not seed- or selection-adjusted uncertainty.

Block-length sensitivity

The stored sensitivity checks use 1,000 replications at mean block lengths 5 and 63. The floored-MSE interval just excludes zero at length 5 but includes it at 21 and 63. The significance conclusion is therefore not stable across these dependence assumptions.

Calendar-period MSE

Period	n	GARCH MSE	Raw MSE	Floored MSE
2010-2016	1739	0.48819	0.45999	0.45972
2017-2019	754	0.27962	0.26222	0.26184
2020-2021	505	8.99661	8.45199	8.45007
2022-2024	740	0.51356	0.52878	0.52863

Table 3. MSE columns are divided by 10^{-7} . Calendar slices are descriptive; all dates remain in the headline comparison. The 2008 crisis is in training history, not the test period.

Concentration of the difference

The floored model has lower MSE in 98 of 178 test blocks. The ten largest absolute daily MSE differences account for 42.36% of total absolute difference. These diagnostics are not independent win-rate tests; no influential dates were removed.

04 / Objective and Forecast Validity

What the additive model learns

$$\begin{aligned}e_t &= y_t - g_t \\h_t^{\text{raw}} &= g_t + \text{predicted } e_t \\(y - h^{\text{raw}})^2 &= (e - \text{predicted } e)^2\end{aligned}$$

The target is a variance-forecast error, not a return residual or standardized GARCH innovation. Training-only affine standardization preserves signed residuals; a linear output minimizes a loss proportional to residual MSE within a fold. This identity aligns the objective with variance MSE but does not ensure generalization or positivity.

Why positive alternatives answer a different question

Direct log-target models exponentiate a log prediction; log-ratio hybrids reconstruct $h = g * \exp(\text{predicted log ratio})$. These transformations ensure positive finite outputs when numerical checks pass, but log-MSE does not generally estimate the arithmetic conditional mean required to minimize variance MSE. Positivity and objective alignment are separate properties.

Fixed-floor sensitivity

Floor	MSE / 10 ⁻⁷	QLIKE	Mean h / mean y
1e-12	1.51293	457,801.0548	1.051
1e-10	1.51293	4,570.4315	1.051
1e-08	1.51293	37.3294	1.051
1e-06	1.51292	-7.9614	1.051

Table 4. The primary floor is 1e-12. The other floors are diagnostics only, not test-selected production settings. Flooring changes the forecast and can strongly affect QLIKE; it does not supply a statistical explanation for near-zero variance.

Research implication

A useful next experiment would combine an explicitly positive variance parameterization with a mean-targeted objective, tested under a fixed multi-seed protocol and new data. That experiment has not been run. The present evidence supports the objective/validity tradeoff, not a deployable improvement.

05 / Code, Evidence, and Scope

Implementation and reproducibility

Repository path	Role
src/data; src/models; src/losses	Chronological features and splits; daily GARCH recursion; recurrent training/reconstruction; loss definitions.
scripts/run_corrected_experiment.py; scripts/run_all_models.py	Prepare the benchmark experiment and train the six log-target/log-ratio neural candidates.
scripts/run_additive_residual.py; scripts/evaluate_additive_residual.py	Train the signed-residual LSTM and score raw/floored outputs on matching dates.
reports/corrected_results.json; reports/additive_results.json	Scores, temporal intervals, diagnostics, input/code/prediction hashes, and environment records.
tests; scripts/smoke_demo.py	Timing, reconstruction, state-recursion, bootstrap, and optional TensorFlow checks; a synthetic offline workflow demo.

The README provides the full command sequence. Build this PDF with reports/build_final_report.py. Its source checks reconcile the shared benchmark scores and parent-artifact hash before combining the evidence. It does not retrain models or change the stored evaluation.

Data access and interpretation limits

Raw market snapshots, full predictions, and training logs are not bundled. Their hashes identify the reviewed artifacts but are not substitutes for access in an independent rerun. Supplying different input bytes creates a different experiment. The offline demo uses synthetic returns; it does not reproduce these historical scores.

The study covers one equity index and one training-seed schedule. Squared returns are noisy variance proxies; repeated inspection creates research-selection risk. Temporal intervals omit seed uncertainty and multiple-comparison adjustment. VIX/gate diagnostics in the supporting artifacts are descriptive, not causal evidence. No trading P&L, execution benefit, multi-asset robustness, or production validation is claimed.

Project provenance

Based on a collaborative research project; this portfolio report is prepared by Eddie Liu.

github.com/SRG-eddieliu/rnn-volatility-lab